BENCH TEST REPORT · SAVED RECORD · OFFLINE REVIEW (HISTORICAL)

Adaptive PLC fault investigation (drive feedback and D1 limits)

CAMPAIGN OBJECTIVE (MUSE PLAN) Execute the evidence-dependent D1 contradictory-limits follow-up and finish Stopped fault-free with no active item/pending completion.

Live PLC evidence | Run completed | Historical record | `portal-20261103131442735-1`

Summary

Generated from the campaign record; no additional model call

- Run completed 2026-11-03 after 3 min 28 s.
- 2 of 2 planned experiments reached a recorded conclusion (2 observed, 0 inconclusive).
- No requirement verdict was recorded; experiments report observations only.
- 2 findings with 31 exact PLC claims; 2 unreviewed.
- Separate review files attached: campaign-audit.json (completed candidate for engineering review); engineering-review.json (pass).

CAMPAIGN EXECUTION

Recorded run status

Run completed

EXPERIMENTS

Experiment conclusions as recorded; observed is not pass

2	Planned
2	Observed
0	Inconclusive
0	Not run

REQUIREMENT VERDICTS

None recorded; experiments report observations only

0	Pass
0	Fail
0	Inconclusive
3	No verdict

FINDINGS

Engineer review state in the campaign record

2	Recorded
2	Unreviewed
0	Accepted

MUSE ACTIONS

Tool calls, counted by outcome

28	Applied
2	Refused
2	Timed out

Needs attention

> 4 Muse actions did not apply: call-start-02 refused, call-wait-ready-02 timed out, call-inject-contra-d1 refused, call-wait-faulted-02 timed out.

> 2 discrepancies recorded outside an active experiment.

> 2 findings awaiting engineer review in the campaign record.

Execution

Completed 2026-11-03 13:18:25.790 UTC.

Muse summary MODEL-AUTHORED · UNREVIEWED

> Completed EXP-FAULT-DRIVEFB-01: PLC output :2016 showed Faulted, FaultCode 12 'Motor running feedback lost', Motor_RunCmd=false, CompletionPending=false; after fault clearance and distinct Reset, :3510 showed Stopped, FaultCode 0, Motor_RunCmd=false, ItemActive=false, CompletionPending=false. Then evidence-dependent follow-up EXP-ADAPT-D1CONTRA-02: PLC output :10359 showed Faulted, FaultCode 21 'D1 contradictory limits', Motor_RunCmd=false, D1_ExtendCmd=false, CompletionPending=false; after clearance and Reset, :11530 showed Stopped, FaultCode 0, Motor_RunCmd=false, ItemActive=false, CompletionPending=false, OfferPending=false. Findings FIND-DRIVEFB-01 and FIND-D1CONTRA-02 recorded with injected truth refs kept separate. No HMI operation was claimed.

Identity and source

Live PLC evidence

Campaign	`portal-20261103131442735-1`
Report title	Adaptive PLC fault investigation (drive feedback and D1 limits)
Machine	*Name not recorded in the campaign record; identified by the specification and I/O map below*
Evidence class	Live PLC evidence
Report source	Saved record · offline review (historical)
Started	2026-11-03 13:14:58.036 UTC
Ended	2026-11-03 13:18:25.790 UTC
Duration	3 min 28 s
Controller build	`8001/interface-1/sha256:6c2a1eb6c9c83485b75cb9bcefc1be237a94c8727ecf8098ad57a4d680516628`
World module	`sha256:277992048f8ca9a8b36f2f90c0d4c33ef479cb8b3c355d6a2233a272850698f5;worldVersion:0.5.0;loaded-by-child:52888`
Specification	`sha256:acd1e39619af301155ac0f8b81f4b824c36c478800b4d482ddb225a2a21d5185`
I/O map	`f8c30539e336f55056f73d4d7ef467b10a9175a60dec3ec156ba1a7bb7a7ba80`
Model	`muse-wonder-1.0`
Connector	`m8-opcua-v1/m8-conveyor-1.0.2;controlsbench.wonder-process.v1;child:9eea397f4ffafcb9`
PLC boot identity	`1751589051509`
Authorized scope	faults, adaptive · known diagnosis

Claim boundary: Wonder-directed M8 campaign through the runtime owner's verified isolated adapter. Scope-limited experiment evidence; no certified controller coverage. HMI frames remain separate.

Test scope and results

2 experiments

Experiment	Requirement refs	Recorded conclusion	Requirement verdict	Evidence
`EXP-FAULT-DRIVEFB-01` Planned	R-11	Observed — conclusion recorded, not a pass verdict	No requirement verdict recorded	`#2016 PLC output` `#3510 PLC output`

Experiment	Requirement refs	Recorded conclusion	Requirement verdict	Evidence
EXP-ADAPT-D1CONTRA-02 Adaptive follow-up (added from evidence)	R-15, R-18	Observed — conclusion recorded, not a pass verdict	No requirement verdict recorded	#10359 PLC output #11530 PLC output

▾ **EXP-FAULT-DRIVEFB-01** faults Observed — conclusion recorded, not a pass verdict

Inject fault.setDriveFeedbackFailure while attempting a carton run; observe whether PLC detects drive-start fault, removes motor command/feeds, and makes no false completion.

Objective	MUSE Inject fault.setDriveFeedbackFailure while attempting a carton run; observe whether PLC detects drive-start fault, removes motor command/feeds, and makes no false completion.
Requirement refs	R-11
Origin	Planned
Run	2026-11-03 13:15:07.825 UTC → 2026-11-03 13:16:34.995 UTC
Stimulus / operator action	• Operator start (direct PLC command, not via HMI) — applied; controller applied #615 Controller action • Inject fault.setDriveFeedbackFailure {"active":true} — applied #1700 Source record #1701 Raw inputs #1702 PLC ack #1703 PLC output • Assign route A to 501 — applied #1724 Controller action • Introduce carton 501 (requested route A) — applied #1741 Source record #1742 Raw inputs #1743 PLC ack #1744 PLC output • Clear fault.setDriveFeedbackFailure {"active":false} — applied #3192 Source record #3193 Raw inputs #3194 PLC ack #3195 PLC output • Operator reset (direct PLC command, not via HMI) — applied; controller applied #3332 Controller action
Expected behaviour	*No explicit acceptance criterion is recorded in the campaign for this experiment.*
Observed behaviour	• Wait for controller.ReleaseReady = true (up to 15000 ms) — observed #657 Observation #650 PLC output • Wait for controller.ModeText = Faulted (up to 30000 ms) — observed #1771 Observation #1768 PLC output • Wait for controller.ModeText = Stopped (up to 15000 ms) — observed #3345 Observation #3342 PLC output

Cited PLC output #2016 PLC output recorded 2026-11-03 13:15:29.200 UTC (source 2026-11-03 13:15:29.066 UTC) · PLC cycle 29155

Motor_RunCmd	false	StackLight_Red	true
controller.ModeText	Faulted	controller.FaultCode	12
controller.FaultText	Motor running feedback lost	controller.ReleaseReady	false
controller.ItemActive	false	controller.OfferPending	true

controller.CompletionPending	false	controller.CountExitA	1

Cited PLC output #3510 PLC output recorded 2026-11-03 13:15:53.547 UTC (source 2026-11-03 13:15:53.533 UTC) · PLC cycle 30359

Motor_RunCmd	false	StackLight_Red	false
controller.ModeText	Stopped	controller.FaultCode	0
controller.FaultText	""	controller.ReleaseReady	false
controller.ItemActive	false	controller.OfferPending	false
controller.CompletionPending	false	controller.CountExitA	1

Recorded conclusion: Observed — conclusion recorded, not a pass verdict

Requirement verdict: No requirement verdict recorded

Supporting evidence: #2016 PLC output #3510 PLC output

Limitations:

- FIND-DRIVEFB-01: Exact drive-feedback detection deadline was not measured from a timestamped fault-onset PLC-output pair in the cited claims; CountExitA=1 is the observed value, baseline comparison is not claimed beyond cited pairs.

▾ EXP-ADAPT-D1CONTRA-02

adaptive | Observed — conclusion recorded, not a pass verdict

From drive-feedback fault evidence (:2016 Faulted/Code12, motor off), test whether D1 contradictory home+extended feedback in Auto produces a distinct PLC fault/indication with no false divert completion.

Objective: MUSE From drive-feedback fault evidence (:2016 Faulted/Code12, motor off), test whether D1 contradictory home+extended feedback in Auto produces a distinct PLC fault/indication with no false divert completion.

Requirement refs: R-15, R-18

Origin: Adaptive follow-up (added from evidence)

MUSE RATIONALE PLC demonstrably faulted on a feedback inconsistency for the drive (FaultCode 12 at :2016) and recovered cleanly to Stopped at :3510; whether divert limit contradiction is similarly diagnosed is untested and not selected before that evidence. #2016 PLC output #3510 PLC output

Run: 2026-11-03 13:17:27.732 UTC → 2026-11-03 13:18:25.779 UTC

Stimulus / operator action:

- Operator start (direct PLC command, not via HMI) — applied; controller applied #9858 Controller action
- Inject fault.setContradictoryLimits {"divert":1,"active":true} — applied #9925 Source record #9926 Raw inputs #9927 PLC ack #9928 PLC output
- Clear fault.setContradictoryLimits {"divert":1,"active":false} — applied #11213 Source record #11214 Raw inputs #11215 PLC ack #11216 PLC output
- Operator reset (direct PLC command, not via HMI) — applied; controller applied #11354 Controller action

Expected behaviour: *No explicit acceptance criterion is recorded in the campaign for this experiment.*

Observed behaviour

- Wait for controller.ReleaseReady = true (up to 15000 ms) — observed `#9903 Observation #9895 PLC output`
- Wait for controller.ModeText = Stopped (up to 15000 ms) — observed `#11367 Observation #11364 PLC output`

Cited PLC output `#10359 PLC output` recorded 2026-11-03 13:17:33.347 UTC (source 2026-11-03 13:17:33.336 UTC) · PLC cycle 35265

Motor_RunCmd	false	D1_ExtendCmd	false
StackLight_Red	true	controller.ModeText	Faulted
controller.FaultCode	21	controller.FaultText	D1 contradictory limits
controller.ReleaseReady	false	controller.ItemActive	false
controller.OfferPending	false	controller.CompletionPending	false
controller.CountExitA	1		

Cited PLC output `#11530 PLC output` recorded 2026-11-03 13:17:52.025 UTC (source 2026-11-03 13:17:52.010 UTC) · PLC cycle 36184

Motor_RunCmd	false	D1_ExtendCmd	false
StackLight_Red	false	controller.ModeText	Stopped
controller.FaultCode	0	controller.FaultText	""
controller.ReleaseReady	false	controller.ItemActive	false
controller.OfferPending	false	controller.CompletionPending	false
controller.CountExitA	1		

Recorded conclusion: Observed — conclusion recorded, not a pass verdict

Requirement verdict: No requirement verdict recorded

Supporting evidence: `#10359 PLC output #11530 PLC output`

Limitations

- FIND-D1CONTRA-02: No carton was introduced in this follow-up, so divert-timing or misroute behavior under an active A assignment was not tested; detection deadline not claimed.

Actions outside an active experiment

- Operator start (direct PLC command, not via HMI) — refused none cited Controller actions require an active authorized experiment
- Wait for controller.ReleaseReady = true (up to 15000 ms) — timed out none cited Observation deadline reached without sufficient matching evidence
- Inject fault.setContradictoryLimits {"divert":1,"active":true} — refused none cited Start an experiment before a world intervention
- Wait for controller.ModeText = Faulted (up to 20000 ms) — timed out none cited Observation deadline reached without sufficient matching evidence

Requirement verdicts

Only verdicts the campaign explicitly recorded; none are inferred from experiment completion

Requirement	Exercised by	Recorded verdict	Separate review artifacts
R-11	EXP-FAULT-DRIVEFB-01 (observed)	No verdict recorded	none attached
R-15	EXP-ADAPT-D1CONTRA-02 (observed)	No verdict recorded	none attached
R-18	EXP-ADAPT-D1CONTRA-02 (observed)	No verdict recorded	none attached

Findings

Recorded by Muse; observed values are exact PLC outputs, inference is unverified

Drive feedback failure faulted PLC with motor command removed and no completion; clear+Reset returned Stopped without restart

FIND-DRIVEFB-01 · Unreviewed

Experiment: EXP-FAULT-DRIVEFB-01

Priority: *Not supplied in the campaign record*

Observed facts: 15 exact PLC claims (✓ = matches the cited ledger record)

At #2016 PLC output recorded 2026-11-03 13:15:29.200 UTC (source 2026-11-03 13:15:29.066 UTC) · PLC cycle 29155

controller.ModeText	Faulted ✓	controller.FaultCode	12 ✓
controller.FaultText	Motor running feedback lost ✓	Motor_RunCmd	false ✓
StackLight_Red	true ✓	controller.ReleaseReady	false ✓
controller.ItemActive	false ✓	controller.CompletionPending	false ✓
controller.CountExitA	1 ✓		

At #3510 PLC output recorded 2026-11-03 13:15:53.547 UTC (source 2026-11-03 13:15:53.533 UTC) · PLC cycle 30359

controller.ModeText	Stopped ✓	controller.FaultCode	0 ✓
Motor_RunCmd	false ✓	controller.ItemActive	false ✓
controller.CompletionPending	false ✓	controller.OfferPending	false ✓

Injected fault (truth record): Injected fault.setDriveFeedbackFailure {"active":true} #1685 Injected fault

Inference MUSE · UNVERIFIED: PLC outputs showed Faulted/FaultCode 12 with motor command removed and no pending completion during injected drive-feedback failure; after clearing the fault and a distinct Reset, outputs showed Stopped/FaultCode 0 with motor command still false, i.e. no auto-restart in the observed window.

Assumptions:

- FaultCode 12 text is taken as PLC-reported string, not inferred from injected truth

Uncertainty	Exact drive-feedback detection deadline was not measured from a timestamped fault-onset PLC-output pair in the cited claims; CountExitA=1 is the observed value, baseline comparison is not claimed beyond cited pairs.
Engineer review	Unreviewed

D1 contradictory limits faulted PLC with distinct code, motor off and no completion; clear+Reset returned Stopped without restart

`FIND-D1CONTRA-02` Unreviewed

Experiment	`EXP-ADAPT-D1CONTRA-02`
Priority	*Not supplied in the campaign record*
Observed facts	16 exact PLC claims (✓ = matches the cited ledger record)

At `#10359 PLC output` recorded 2026-11-03 13:17:33.347 UTC (source 2026-11-03 13:17:33.336 UTC) · PLC cycle 35265

`controller.ModeText`	`Faulted` ✓	`controller.FaultCode`	`21` ✓
`controller.FaultText`	`D1 contradictory limits` ✓	`Motor_RunCmd`	`false` ✓
`D1_ExtendCmd`	`false` ✓	`StackLight_Red`	`true` ✓
`controller.ReleaseReady`	`false` ✓	`controller.ItemActive`	`false` ✓
`controller.CompletionPending`	`false` ✓	`controller.CountExitA`	`1` ✓

At `#11530 PLC output` recorded 2026-11-03 13:17:52.025 UTC (source 2026-11-03 13:17:52.010 UTC) · PLC cycle 36184

`controller.ModeText`	`Stopped` ✓	`controller.FaultCode`	`0` ✓
`Motor_RunCmd`	`false` ✓	`controller.ItemActive`	`false` ✓
`controller.CompletionPending`	`false` ✓	`controller.OfferPending`	`false` ✓

Injected fault (truth record)	Injected fault.setContradictoryLimits {"divert":1,"active":true} `#9907 Injected fault`
Inference MUSE · UNVERIFIED	In contrast to drive-feedback FaultCode 12 at :2016 in the prior experiment, PLC outputs for the D1 contradictory-limits condition showed a distinct Faulted/FaultCode 21 with motor and D1 commands false and no pending completion; after clearing and a distinct Reset, outputs showed Stopped/FaultCode 0 with motor command still false.
Assumptions	• FaultCode 21 text is taken as PLC-reported string, not inferred from injected truth
Uncertainty	No carton was introduced in this follow-up, so divert-timing or misroute behavior under an active A assignment was not tested; detection deadline not claimed.
Engineer review	Unreviewed

HMI behaviour

This campaign record contains no HMI actions, HMI captures or HMI assessments.

No HMI behaviour was observed or assessed in this campaign record.

Recommended next checks

RECOMMENDATIONS GENERATED FROM GAPS IN THIS RECORD · NOT TEST RESULTS

1. Review the 4 actions that did not apply before relying on the affected sequence.
2. Define acceptance criteria for R-11, R-15, R-18 so observations can become requirement verdicts.
3. Engineer review: accept or request a rerun for each unreviewed finding and assessment.

Remaining work, missing evidence and limits

Actions that did not apply

- call-start-02 (operator action) refused: Controller actions require an active authorized experiment
- call-wait-ready-02 (wait for condition) timed out: Observation deadline reached without sufficient matching evidence
- call-inject-contra-d1 (inject fault) refused: Start an experiment before a world intervention
- call-wait-faulted-02 (wait for condition) timed out: Observation deadline reached without sufficient matching evidence

Scope limits

- Wonder-directed M8 campaign through the runtime owner's verified isolated adapter. Scope-limited experiment evidence; no certified controller coverage. HMI frames remain separate.
- Experiment evidence only; no final conveyor-controller requirement coverage is certified.
- Expected PLC condition was not observed within the bounded window (expected controller.ReleaseReady = true) #7107
- Expected PLC condition was not observed within the bounded window (expected controller.ModeText = Faulted) #8769

Final recorded PLC and connection state

historical

Last PLC output recorded in this campaign, at 2026-11-03 13:18:25.846 UTC. Historical: this report does not monitor current PLC state or freshness.

PLC output recorded	2026-11-03 13:18:25.846 UTC (source 2026-11-03 13:18:25.831 UTC)
PLC cycle	37848

PLC boot identity	1751589051509
Connector state	fresh — Observed simulation active under stable PLC boot (recorded 2026-11-03 13:18:25.832 UTC)

▼ All 40 recorded output values

Motor_RunCmd	false	D1_ExtendCmd	false
D2_ExtendCmd	false	StackLight_Red	false
StackLight_Green	false	controller.Mode	0
controller.ModeText	Stopped	controller.FaultCode	0
controller.FaultText	""	controller.ExceptionCode	0
controller.ExceptionText	""	controller.ReleaseReady	false
controller.ItemActive	false	controller.ActiveItemId	0
controller.RequestedRoute	0	controller.EffectiveRoute	0
controller.ActualExit	3	controller.Disposition	2
controller.OfferAckSequence	2	controller.OfferAckItemId	501
controller.OfferAccepted	true	controller.OfferReason	0
controller.OfferPending	false	controller.CompletionSequence	2
controller.CompletionPending	false	controller.CompletionItemId	0
controller.CompletionRoute	0	controller.CompletionExit	3
controller.CompletionDisposition	2	controller.CountExitA	1
controller.CountExitB	0	controller.CountRunout	1
controller.CountMisroute	0	controller.CountFaulted	0
controller.StartAck	3	controller.StopAck	0
controller.ResetAck	4	controller.StartAccepted	true
controller.ResetAccepted	true	controller.EncoderDelta	0

Usage and accounting

Reservations and estimates, not invoices

Budget reservations

This campaign	$45.00
Session total (includes earlier campaigns)	$105.00
Held for planning	$10.00
Held for EXP-FAULT-DRIVEFB-01	$20.00
Held for EXP-ADAPT-D1CONTRA-02	$15.00

Conservative reservations are admission accounting held before each model request. They are not spend and not an invoice.

Model token usage (as reported per request)

Model requests	9 (9 completed)
Input tokens	831,745
of which cached	514,604
Output tokens	8,850
of which reasoning	3,855

Token-cost estimates

Estimate at uncached list rates (9 of 9 requests priced)	≈ $3.5040
Estimate stated in engineering-review.json	≈ $3.5040

Token-cost estimates use the pricing recorded with each request (4/20 USD per million input/output tokens, checked 2026-11-01). Estimates, not the invoice; caching and billing can differ.

Separate review artifacts

Not part of the model report; status and origin shown as recorded

▾ **`campaign-audit.json`** offline audit · status: completed candidate for engineering review

Separate offline audit of the saved evidence. Not part of the model report; it does not change any recorded conclusion.

Recorded	2026-11-03 13:19:54.892 UTC
File SHA-256	`47ad324e4a363b18f04699d478f95015200e14610d9164fbe325005662c39ffb`
Report reconstructed exactly from ledger	yes
Original evidence hashes verified	7
Model response hashes verified	9
Response-to-tool lineage verified	yes
Exact paired terminal results	31
Model requests	9
Completed responses	9
Recorded failures	none
Service retries	0

Observed PLC fault codes	12, 21
Claim boundary	Audit proves preserved evidence and exact claims. Review follow-up novelty, actual interventions and inferences before calling W4 passed. No agent HMI-operation or certified coverage claim. Known reported token usage only at documented uncached rates; actual invoices/caching differ. Failed and unreturned generations are not assumed free.

▾ **`engineering-review.json`** engineering review status: pass

Separate engineering review artifact. Not part of the model report; the original Muse findings and assessments remain unreviewed in the campaign record.

Scope	live-m8-fault-and-evidence-dependent-physical-followup
Recorded	2026-11-03 13:21:33.819 UTC
File SHA-256	`cf9b274cc268b407bf20ad5f5c5da9f38b1f823daf4e362899cca6e18dc76a44`
Claims traced to original child/transport records	31
Model requests	9
Model service retries	0
Prior normal operation	Historical normal-sort proof is labelled separately in the packet; it was not repeated in this campaign.
Follow-up novelty	The initial plan contained only drive-feedback failure. After observing fault12 and clean recovery, Muse proposed a distinct D1-limit contradiction test, explicitly generalizing the observed feedback-inconsistency response. Its cited current observations predate the proposal. This is a modest evidence-dependent extension, not discovery of an unknown fault.
Execution	Start, contradictory-limit injection, clear and Reset all executed in the newly active follow-up; native receipts and original child observations support both distinct fault21 and recovery. No human intervention occurred during the campaign.
Claim boundary	Scoped fault/adaptive checkpoint passed; not a repeat of normal sorting, HMI-agent proof, exhaustive controller validation or general machine coverage. Known reported token usage only at documented uncached rates; actual invoices/caching differ. Failed and unreturned generations are not assumed free.

Limitations stated by the review

- The first proposed same-turn follow-up actions were refused before activation; observation waits timed out. Muse corrected this in the next turn. Those attempts remain in the evidence.
- No carton was introduced for the D1 follow-up; misrouting and divert timing were not tested.
- The first fault was running-feedback loss (12), not the startup-feedback deadline. Detection deadlines are not established.

- The world retains one stationary virtual carton while the PLC has no tracked item. No world cleanup is represented as physical travel.
- The new confirmed-PLC-rejection handling passed offline tests but was not exercised by a PLC rejection in this successful run.
- Muse HMI operation remains a later checkpoint; these were semantic engine/controller tools.

Evidence appendix

69 cited records of 13,464 ledger events; the full ledger is a separate file

Integrity and sources

Report	`report.json sha256 db7964a397e0e4ab9056c73a6b5a8cab9e6e2c7fcfcb8ea5401d7418522ae91e`
Matches reconstruction from ledger	yes
Ledger	`13,464 events · head ac2b2d872ce8e975… · tail e29c0742ac707b3c…`
Recorded span	2026-11-03 13:14:43.029 UTC → 2026-11-03 13:18:25.876 UTC
Report generated	2026-11-04 00:37:45.798 UTC

Source file	Role	SHA-256	Size
`report.json`	Original structured campaign report (unchanged)	`db7964a397e0e4ab9056c73a6b5a8cab9e6e2c7fcfcb8ea5401d7418522ae91e`	227.7 KiB
`campaign.jsonl`	Full hash-chained campaign ledger · 13,464 events	`bea3a3a68987bcd3003a424f121123aea5637a4fa1b27968930c0439738bd25f`	17185.0 KiB
`campaign-audit.json`	Separate offline audit	`47ad324e4a363b18f04699d478f95015200e14610d9164fbe325005662c39ffb`	10.1 KiB
`engineering-review.json`	Separate engineering review	`cf9b274cc268b407bf20ad5f5c5da9f38b1f823daf4e362899cca6e18dc76a44`	2.4 KiB
`wonder-exchanges.jsonl`	Model request/response and image-input lineage	`7345f32926082bf390f4960747a6df7b0363e200a3794345767bc671cae43ddf`	919.7 KiB

▼ Muse decision trail (9 model requests; decisions are model-authored)

Request	Started · elapsed	Tokens	Decision summary (Muse)
request-1 completed planning	2026-11-03 13:14:58.050 UTC 10 s	15,518 in (0 cached) · 990 out response returned · 0 image inputs	Plan drive-feedback fault experiment and start PLC to observe baseline Auto/ReleaseReady before fault injection.
request-2 completed EXP-FAULT-DRIVEFB-01	2026-11-03 13:15:10.432 UTC 15 s	36,395 in (0 cached) · 592 out response returned · 0 image inputs	In EXP-FAULT-DRIVEFB-01 baseline Auto+ReleaseReady observed (evidence 788/824). Now inject drive feedback failure, assign A to item 501, introduce it, and wait for Faulted mode.
request-3 completed EXP-FAULT-DRIVEFB-01	2026-11-03 13:15:29.229 UTC 21 s	55,193 in (36,340 cached) · 1,169 out response returned · 0 image inputs	Faulted evidence (:2016 FaultCode=12, Motor_RunCmd=false, CompletionPending=false) obtained. Now clear drive feedback fault and Reset to test recovery/no-auto-restart within EXP-FAULT-DRIVEFB-01.
request-4 completed EXP-FAULT-DRIVEFB-01	2026-11-03 13:15:53.595 UTC 21 s	73,895 in (15,476 cached) · 1,818 out response returned · 0 image inputs	Record finding for EXP-FAULT-DRIVEFB-01 from faulted evidence :2016 and stopped-after-reset evidence :3510, with injected truth ref :1685 kept separate.
request-5 completed EXP-FAULT-DRIVEFB-01	2026-11-03 13:16:14.955 UTC 20 s	92,404 in (55,156 cached) · 1,352 out response returned · 0 image inputs	Complete EXP-FAULT-DRIVEFB-01 (observed: Faulted FaultCode12 at :2016, Stopped FaultCode0 after clear+Reset at :3510), then create and start a new follow-up probing D1 contradictory limits feedback inconsistency.

Request	Started · elapsed	Tokens	Decision summary (Muse)
request-6 completed planning	2026-11-03 13:17:13.092 UTC 15 s	111,235 in (92,404 cached) · 810 out response returned · 0 image inputs	Activate proposed EXP-ADAPT-D1CONTRA-02 with its exact definition, then Start, inject D1 contradictory limits, and observe PLC response.
request-7 completed EXP-ADAPT-D1CONTRA-02	2026-11-03 13:17:33.475 UTC 15 s	130,199 in (73,844 cached) · 419 out response returned · 0 image inputs	Clear D1 contradictory-limits fault (:10359 Faulted/Code21) and Reset, then observe stopped fault-free state for EXP-ADAPT-D1CONTRA-02.
request-8 completed EXP-ADAPT-D1CONTRA-02	2026-11-03 13:17:52.118 UTC 15 s	149,079 in (130,164 cached) · 1,160 out response returned · 0 image inputs	Record finding for EXP-ADAPT-D1CONTRA-02 from faulted evidence :10359 and stopped-after-reset evidence :11530, injected truth :9907 kept separate.
request-9 completed EXP-ADAPT-D1CONTRA-02	2026-11-03 13:18:07.349 UTC 18 s	167,827 in (111,220 cached) · 540 out response returned · 0 image inputs	Complete EXP-ADAPT-D1CONTRA-02 from faulted evidence :10359 and stopped evidence :11530, then finish campaign. Final observed state at :11530 was Stopped, FaultCode 0, Motor_RunCmd false, ItemActive false, CompletionPending false and OfferPending false.

Cited evidence records

▼ Evidence #615 – #9927 (50 records)

Evidence	Recorded	Summary
#615 Controller action	2026-11-03 13:15:07.968 UTC	Controller start applied portal-20261103131442735-1:615

Evidence	Recorded	Summary
#650 PLC output	2026-11-03 13:15:08.375 UTC source 2026-11-03 13:15:08.359 UTC	PLC outputs at cycle 28136 cycle 28136 · boot 1751589051509 ▶ 40 values portal-20261103131442735-1:650
#657 Observation	2026-11-03 13:15:08.402 UTC	Observed controller.ReleaseReady = true portal-20261103131442735-1:657
#667 PLC output	2026-11-03 13:15:08.490 UTC source 2026-11-03 13:15:08.474 UTC	PLC outputs at cycle 28142 cycle 28142 · boot 1751589051509 ▶ 40 values portal-20261103131442735-1:667
#824 PLC output	2026-11-03 13:15:10.331 UTC source 2026-11-03 13:15:10.314 UTC	PLC outputs at cycle 28233 cycle 28233 · boot 1751589051509 ▶ 40 values portal-20261103131442735-1:824
#828 Observation	2026-11-03 13:15:10.421 UTC	Observation window 2000 ms completed portal-20261103131442735-1:828
#1685 Injected fault	2026-11-03 13:15:25.470 UTC	Injected fault.setDriveFeedbackFailure {"active":true} portal-20261103131442735-1:1685
#1700 Source record	2026-11-03 13:15:25.618 UTC	Coordinator connector commit.acknowledged portal-20261103131442735-1:1700
#1701 Raw inputs	2026-11-03 13:15:25.620 UTC source 2026-11-03 13:15:25.609 UTC	Raw input image, commit sequence 255 cycle 28985 · boot 1751589051509 ▶ 20 values portal-20261103131442735-1:1701
#1702 PLC ack	2026-11-03 13:15:25.621 UTC	PLC applied commit 255 (46.8 ms) portal-20261103131442735-1:1702
#1703 PLC output	2026-11-03 13:15:25.623 UTC source 2026-11-03 13:15:25.609 UTC	PLC outputs at cycle 28985 cycle 28985 · boot 1751589051509 ▶ 40 values portal-20261103131442735-1:1703
#1724 Controller action	2026-11-03 13:15:25.751 UTC	Controller offer applied portal-20261103131442735-1:1724
#1741 Source record	2026-11-03 13:15:25.940 UTC	Coordinator connector commit.acknowledged portal-20261103131442735-1:1741

Evidence	Recorded	Summary

#1742
Raw inputs
	2026-11-03 13:15:25.943 UTC
	source 2026-11-03 13:15:25.932 UTC
		Raw input image, commit sequence 258
		`cycle 29001 · boot 1751589051509`
		▶ 20 values
		`portal-20261103131442735-1:1742`

#1743 PLC ack	2026-11-03 13:15:25.945 UTC	PLC applied commit 258 (40.6 ms) `portal-20261103131442735-1:1743`

#1744
PLC output
	2026-11-03 13:15:25.946 UTC
	source 2026-11-03 13:15:25.932 UTC
		PLC outputs at cycle 29001
		`cycle 29001 · boot 1751589051509`
		▶ 40 values
		`portal-20261103131442735-1:1744`

#1768
PLC output
	2026-11-03 13:15:26.193 UTC
	source 2026-11-03 13:15:26.181 UTC
		PLC outputs at cycle 29013
		`cycle 29013 · boot 1751589051509`
		▶ 40 values
		`portal-20261103131442735-1:1768`

#1771 Observation	2026-11-03 13:15:26.202 UTC	Observed controller.ModeText = Faulted `portal-20261103131442735-1:1771`

#1782
PLC output
	2026-11-03 13:15:26.273 UTC
	source 2026-11-03 13:15:26.259 UTC
		PLC outputs at cycle 29017
		`cycle 29017 · boot 1751589051509`
		▶ 40 values
		`portal-20261103131442735-1:1782`

#2016
PLC output
	2026-11-03 13:15:29.200 UTC
	source 2026-11-03 13:15:29.066 UTC
		PLC outputs at cycle 29155
		`cycle 29155 · boot 1751589051509`
		▶ 40 values
		`portal-20261103131442735-1:2016`

#2021 Observation	2026-11-03 13:15:29.218 UTC	Observation window 3000 ms completed `portal-20261103131442735-1:2021`
#3192 Source record	2026-11-03 13:15:49.910 UTC	Coordinator connector commit.acknowledged `portal-20261103131442735-1:3192`

#3193
Raw inputs
	2026-11-03 13:15:49.912 UTC
	source 2026-11-03 13:15:49.896 UTC
		Raw input image, commit sequence 479
		`cycle 30180 · boot 1751589051509`
		▶ 20 values
		`portal-20261103131442735-1:3193`

#3194 PLC ack	2026-11-03 13:15:49.915 UTC	PLC applied commit 479 (45.2 ms) `portal-20261103131442735-1:3194`

Evidence	Recorded	Summary
#3195 PLC output	2026-11-03 13:15:49.918 UTC source 2026-11-03 13:15:49.896 UTC	PLC outputs at cycle 30180 cycle 30180 · boot 1751589051509 ▶ 40 values portal-20261103131442735-1:3195
#3209 PLC output	2026-11-03 13:15:50.013 UTC source 2026-11-03 13:15:49.994 UTC	PLC outputs at cycle 30185 cycle 30185 · boot 1751589051509 ▶ 40 values portal-20261103131442735-1:3209
#3326 PLC output	2026-11-03 13:15:51.435 UTC source 2026-11-03 13:15:51.418 UTC	PLC outputs at cycle 30255 cycle 30255 · boot 1751589051509 ▶ 40 values portal-20261103131442735-1:3326
#3329 Observation	2026-11-03 13:15:51.444 UTC	Observation window 1500 ms completed portal-20261103131442735-1:3329
#3332 Controller action	2026-11-03 13:15:51.482 UTC	Controller reset applied portal-20261103131442735-1:3332
#3342 PLC output	2026-11-03 13:15:51.556 UTC source 2026-11-03 13:15:51.541 UTC	PLC outputs at cycle 30261 cycle 30261 · boot 1751589051509 ▶ 40 values portal-20261103131442735-1:3342
#3345 Observation	2026-11-03 13:15:51.565 UTC	Observed controller.ModeText = Stopped portal-20261103131442735-1:3345
#3355 PLC output	2026-11-03 13:15:51.679 UTC source 2026-11-03 13:15:51.658 UTC	PLC outputs at cycle 30267 cycle 30267 · boot 1751589051509 ▶ 40 values portal-20261103131442735-1:3355
#3510 PLC output	2026-11-03 13:15:53.547 UTC source 2026-11-03 13:15:53.533 UTC	PLC outputs at cycle 30359 cycle 30359 · boot 1751589051509 ▶ 40 values portal-20261103131442735-1:3510
#3514 Observation	2026-11-03 13:15:53.587 UTC	Observation window 2000 ms completed portal-20261103131442735-1:3514
#4720 finding.created	2026-11-03 13:16:14.939 UTC	Finding FIND-DRIVEFB-01 portal-20261103131442735-1:4720

Evidence	Recorded	Summary
#5856 Follow-up	2026-11-03 13:16:35.008 UTC	Follow-up EXP-ADAPT-D1CONTRA-02 added `portal-20261103131442735-1:5856`

#7107
PLC output
	2026-11-03 13:16:50.022 UTC
	source 2026-11-03 13:16:50.005 UTC
		PLC outputs at cycle 33135
		`cycle 33135 · boot 1751589051509`
		▶ 40 values
		`portal-20261103131442735-1:7107`

#7110 Discrepancy	2026-11-03 13:16:50.039 UTC	Expected PLC condition was not observed within the bounded window `portal-20261103131442735-1:7110`

#8769
PLC output
	2026-11-03 13:17:09.975 UTC
	source 2026-11-03 13:17:09.962 UTC
		PLC outputs at cycle 34116
		`cycle 34116 · boot 1751589051509`
		▶ 40 values
		`portal-20261103131442735-1:8769`

#8773 Discrepancy	2026-11-03 13:17:10.067 UTC	Expected PLC condition was not observed within the bounded window `portal-20261103131442735-1:8773`

#8782
PLC output
	2026-11-03 13:17:10.099 UTC
	source 2026-11-03 13:17:10.086 UTC
		PLC outputs at cycle 34122
		`cycle 34122 · boot 1751589051509`
		▶ 40 values
		`portal-20261103131442735-1:8782`

#9025
PLC output
	2026-11-03 13:17:13.032 UTC
	source 2026-11-03 13:17:13.020 UTC
		PLC outputs at cycle 34266
		`cycle 34266 · boot 1751589051509`
		▶ 40 values
		`portal-20261103131442735-1:9025`

#9029 Observation	2026-11-03 13:17:13.082 UTC	Observation window 3000 ms completed `portal-20261103131442735-1:9029`
#9858 Controller action	2026-11-03 13:17:27.769 UTC	Controller start applied `portal-20261103131442735-1:9858`

#9895
PLC output
	2026-11-03 13:17:28.153 UTC
	source 2026-11-03 13:17:28.140 UTC
		PLC outputs at cycle 35009
		`cycle 35009 · boot 1751589051509`
		▶ 40 values
		`portal-20261103131442735-1:9895`

#9903 Observation	2026-11-03 13:17:28.174 UTC	Observed controller.ReleaseReady = true `portal-20261103131442735-1:9903`
#9907 Injected fault	2026-11-03 13:17:28.183 UTC	Injected fault.setContradictoryLimits {"divert":1,"active":true} `portal-20261103131442735-1:9907`

Evidence	Recorded	Summary
#9925 Source record	2026-11-03 13:17:28.351 UTC	Coordinator connector commit.acknowledged `portal-20261103131442735-1:9925`

#9926
Raw inputs
	2026-11-03 13:17:28.352 UTC
	source 2026-11-03 13:17:28.342 UTC
		Raw input image, commit sequence 1385
		`cycle 35019 · boot 1751589051509`
		▶ 20 values
		`portal-20261103131442735-1:9926`

#9927 PLC ack	2026-11-03 13:17:28.354 UTC	PLC applied commit 1385 (40.0 ms) `portal-20261103131442735-1:9927`

▶ Evidence #9928 – #13461 (19 records)

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